One of the tracks in Figure 3(*b*) shows a sharp deflection. Obviously, something out of the ordinary has happened here to the alpha particle. It has come into the vicinity of the nucleus of an atom and been deflected by it. But, as a rule, the particles move in a straight line, and their range varies mostly between 2 and 10 centimetres.

It is very remarkable that the alpha particles are capable of covering such great distances in a straight line. For it is easy to calculate that on their way they must encounter an enormous number of atoms. This is an obvious inference from the large number of water droplets formed. It seems therefore as if the atoms were not impermeable to particles of such small size, as if, in fact, the latter were able to penetrate the atoms without hindrance. Evidently, they seldom encounter any serious obstacle on their way; but when they do, their tracks will show such deflections, as the one which we see in Figure 3 (*b*).

At an even earlier date, Lenard had investigated the passage of rapidly moving electrons through matter, and discovered that electrons were capable of penetrating extraordinarily. thick layers of matter. Thus he reached the conclusion that the space occupied by an atom is, mostly, empty, and the path of an electron is influenced by individual centres of force only, which he called *dynamides.* It was Rutherford who, as the result of similar studies, took the important step which eventually led to the construction of the first atom model. He studied the tracks of alpha particles through thin metal foil, and concluded, on the ground of the extreme rarity of any notable deflection, that only a very small part of the atom offered any resistance at all to the alpha particles, and that this very small part contained practically all of the mass of the atom. If this were not the case, the laws of elastic impacts would make the occurrence of those considerable deflections, which could at times be observed, quite impossible. On the basis of his observations, Rutherford was able to calculate the volume of the space actually occupied by mass, and concluded that all the rest of the space occupied by the atom must—for all practical purposes—be empty. Rutherford’s collaborators, Geiger and Marsden, succeeded in establishing, furthermore, that the deflections of the positively charged alpha particles were produced by electric forces, due to a likewise positive charge on the central part of the atom. Evidently, this central part of the atom and the alpha particles mutually repelled each other, in conformity with Coulomb’s well-known law.